

Solution Requirements

Date: 29 September 2026 | Team ID: 6ab7959e4c4a9ef0de66bbbee | Project Name: AI FitTrack API | Maximum Marks: 4 Marks

Functional Requirements (FR) — what the AI FitTrack API should do.

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The API should support secure user authentication using JWT-based access control.	High
2	Authorization levels	The API should validate authenticated access and restrict protected API operations to authorized requests. Specific user roles are not defined in the source configuration.	High
3	External interfaces	The API should connect with the configured Gemini AI service for AI-assisted processing and use MongoDB for application data storage.	High
4	Transactions processing	The API should receive requests, validate/route them, process the required operation, and return an API response. No payment workflow is specified in the source.	High
5	Reporting	No dedicated reporting, dashboard, or export functionality is specified in the source configuration; therefore no additional reporting requirement is assumed.	Low
6	Business rules, etc.	Core rules should include request validation/routing, JWT-based secure access, interaction with configured AI functionality, and persistence through MongoDB where required.	High
7	Compliance to laws or regulations	The source configuration does not specify a particular legal or regulatory standard. Data handling should therefore follow applicable requirements, but no specific compliance claim is assumed.	Medium
8	Other	The API should expose the configured backend service through the defined application port and return responses to client requests.	Medium

Non-Functional Requirements (NFR) — how the AI FitTrack API should operate.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The API should process requests and return responses efficiently. The source does not define a numerical response-time target.	No fixed response-time metric specified in source.
2	Scalability	The backend should be structured so the API can handle growth in requests and stored application data.	No specific concurrent-user or data-volume target specified in source.
3	Security & Data Privacy	Use JWT-based authentication for secure access and protect configured connection credentials and API secrets from exposure.	Authenticated access is required; exact encryption standard is not specified in source.
4	Reliability & Availability	The API should remain available on its configured backend service and maintain connectivity with its configured data/AI services.	No uptime percentage or fault-tolerance target specified in source.
5	Usability & Accessibility	API responses should be clear and usable by the client/application consuming the API.	No specific accessibility standard is specified in source.
6	Other	Maintainability should be supported through configuration-based service settings such as the application port, database connection, authentication secret, and AI model configuration.	Configuration values should be managed separately from application logic; no formal maintenance metric specified.

Note: Requirements above are grounded in the provided source configuration and template. Where the source does not specify a feature, role, metric, or compliance standard, it is explicitly marked rather than invented.